L7a: Utility-Based Allocation and the Adaptive Rebalancing Engine

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Objectives for Today

Today we stop treating a portfolio as a vector of weights chosen once and treat it as a process: preferences computed every day from the single index model and the state of the market, answering which assets belong in the basket and how much of each, and a rebalancing engine with an account, rules, and a scorecard.

Today's concepts

- Why a static allocation fails: how buy-and-hold weights drift with prices and how to measure it, and the three ways the minimum-variance inputs let the optimizer down.
- Utility-based allocation: the budget-constrained problem, the Cobb–Douglas closed form, CES and its three limits, and preference weights from the SIM and two market inputs: signs choose the basket, magnitudes the weights.
- The rebalancing engine: a self-financing account with costs, next-bar execution, three trigger rules, and how to score a realized path.

Lectures and Examples

Lecture:
CHEME-5660-L7a-Lecture-Utility-Allocation-Rebalancing-Fall-2026.ipynb

Example 1:
CHEME-5660-L7a-Example-Utility-Allocator-Fall-2026.ipynb

Example 2:
CHEME-5660-L7a-Example-Adaptive-Rebalancing-Scorecard-Fall-2026.ipynb

Example 3:
CHEME-5660-L7a-Example-Portfolio-Drift-Fall-2026.ipynb

The first example builds the allocator and reads which firms belong in the basket; the second runs the engine through 2025 against buy-and-hold baselines; the third measures the drift of a tangent portfolio held through the year.

Concept Review: Allocation as a Time-Stamped Decision

L5b and L6b chose weights $\mathbf{w}$ over $|\mathcal{P}|$ risky assets from an expected growth-rate vector and a growth-rate covariance; with the SIM of L6a those inputs are $\hat{\mathbf{g}}_{\mathrm{SIM}} = \hat{\alpha} + \hat{\beta}\, g'_M$ and $\hat{\Sigma}_{g,\mathrm{SIM}} = s^2_{g,M}\hat{\beta}\hat{\beta}^\top + \hat{\mathbf{D}}_g$ (g'_M, $s^2_{g,M}$ the training-period market mean and variance; $\hat{\mathbf{D}}_g$ the diagonal matrix of residual variances).

- Target $g_\star$: $\mathbf{w}(g_\star)$ minimizes $\mathbf{w}^\top\hat{\Sigma}_g\mathbf{w}$ subject to $\mathbf{w}^\top\hat{\mathbf{g}} = g_\star$, $\mathbf{1}^\top\mathbf{w} = 1$, $w_i \geq 0$; the lowest point is $\mathbf{w}_{\mathrm{GMV}}$; with a risk-free asset earning g_f, the fully invested point of the ray is the tangent portfolio $\mathbf{w}_{\mathcal{T}}$.
- Implemented on the decision date $t = 0$ by buying share counts, and held:

$$n_i = \frac{w_i\, W_{\mathcal{P}}(0)}{S_i(0)}, \qquad W_{\mathcal{P}}(t) = \sum_{i\in\mathcal{P}} n_i\, S_i(t)$$

Two things to say out loud: the weights are optimal on the decision date, for the inputs estimated on that date; and the only decision the investor ever made was the one at $t = 0$. Today is about the other days.

Why Static Allocation Fails: Mechanical Drift

With fixed share counts (no cash, no dividends), $w_i(t) = n_i S_i(t)/W_{\mathcal{P}}(t)$; with $n_i = w_i(0) W_{\mathcal{P}}(0)/S_i(0)$ the initial wealth cancels:

$$w_i(t) = \frac{w_i(0)\ \overbrace{\big(S_i(t)/S_i(0)\big)}^{\text{price relative}}}{\sum_{j\in\mathcal{P}} w_j(0)\big(S_j(t)/S_j(0)\big)}$$

Weights stay put only if every held asset has the same price relative, which does not happen in practice. The drift on day t is the half-L^1 distance, the fraction of wealth a rebalance back to $w_i(0)$ would trade (before costs):

$$\frac{1}{2}\sum_{i\in\mathcal{P}} |w_i(t) - w_i(0)|$$

Example 3 (SIM tangent portfolio through 2025: drift 4.6% by year end):

CHEME-5660-L7a-Example-Portfolio-Drift-Fall-2026.ipynb

Why Static Allocation Fails: Fragile Inputs

Drift is the benign failure; the optimizer's inputs are the serious one. Three tendencies of estimated minimum-variance and tangent portfolios recur:

- Input sensitivity: estimated inputs are treated as exact, so estimation errors move the weights; the tangent portfolio overweights assets whose growth is overstated (Michaud's *error maximizer*, 1989), and errors in the means matter far more than errors in the variances (Chopra and Ziemba, 1993).
- Concentration: without binding constraints the tangent portfolio, and often the GMV portfolio, holds a few names (four of thirteen in L6b).
- Assumption fragility: one stable distribution is assumed, but correlations rise and volatilities jump in the episodes that matter.

Two responses: change the objective, writing preferences for each asset explicitly and letting them respond to the market state (today); or update the inputs as data arrive (L7b).

Utility Maximization

Instead of the smallest variance for a target, ask a consumer-choice question: given a budget and today's market, which portfolio does the investor prefer most? A utility function ranks portfolios, larger for more preferred. At time t, with prices $S_i(t)$ (USD per share) and wealth $W_{\mathcal{P}}(t)$ (USD), choose the share counts $n_i \geq 0$:

$$\boxed{\underset{n_1,\dots,n_{|\mathcal{P}|}}{\text{maximize}} \; U(n_1,\dots,n_{|\mathcal{P}|}) \qquad \text{subject to} \qquad \sum_{i\in\mathcal{P}} n_i\, S_i(t) = W_{\mathcal{P}}(t)}$$

- A holding utility: a preference over the shares held today at today's prices, not the expected utility of an uncertain terminal wealth; risk enters through the preferences and the engine's rules.
- Two coupled questions: which assets belong in the basket today, and how much of each. The utility functions answer the second; the preference model answers the first. Both are recomputed every trading day.

Cobb–Douglas Utility: Sign Classifies, Magnitude Allocates

Let $\gamma_i \in (-1, 1)$ be the preference weight of asset i. Its sign splits the assets into the preferred set $\mathcal{A}^+ = \{i : \gamma_i > 0\}$, the basket, and the non-preferred set $\mathcal{A}^- = \{i : \gamma_i \leq 0\}$, held at a floor of $n_{\min} \geq 0$ shares each; the utility is a product of powers over the preferred set, $U = \prod_{i \in \mathcal{A}^+} n_i^{\gamma_i}$ ($n_i > 0$ on $\mathcal{A}^+$, $n_i = n_{\min}$ on $\mathcal{A}^-$). With $W_{\text{adj}} = W_{\mathcal{P}}(t) - n_{\min} \sum_{k \in \mathcal{A}^-} S_k(t) > 0$ the budget left after the floors and at least one preferred asset, the maximizer is closed form:

$$\boxed{n_i^\star = \underbrace{\frac{\gamma_i}{\sum_{j \in \mathcal{A}^+} \gamma_j}}_{\text{preference share}} \cdot \underbrace{\frac{W_{\text{adj}}}{S_i(t)}}_{\text{budget in shares}} \qquad (i \in \mathcal{A}^+), \qquad n_i^\star = n_{\min} \quad (i \in \mathcal{A}^-)}$$

The dollar weight of a preferred asset within the preferred budget is its normalized preference weight $\gamma_i / \sum_{j \in \mathcal{A}^+} \gamma_j$, independent of prices (times $W_{\text{adj}}/W_{\mathcal{P}}(t)$ as a fraction of total wealth). An empty basket means floors plus cash.

Cobb–Douglas Utility: Derivation

Maximizing U over $\mathcal{A}^+$ is maximizing $\ln U = \sum_{i \in \mathcal{A}^+} \gamma_i \ln n_i$ (the log is increasing). With multiplier λ on the budget over the preferred set, the first-order condition for asset i is:

$$\frac{\gamma_i}{n_i} = \lambda S_i(t) \qquad \Longleftrightarrow \qquad \underbrace{n_i S_i(t)}_{\text{dollars in } i} = \frac{\gamma_i}{\lambda}$$

Each preferred asset receives dollars in proportion to its exponent. Summing over $\mathcal{A}^+$ gives $W_{\text{adj}} = \sum_{j \in \mathcal{A}^+} \gamma_j / \lambda$, and substituting back gives the boxed result; strict concavity of $\ln U$ on the positive orthant makes it the unique maximum.

- The closed form applies to the preferred set only: a non-positive γ_i is a classification, not an exponent to insert (for $\gamma_i < 0$ the term $n_i^{\gamma_i}$ blows up as $n_i \downarrow 0$).
- The allocation responds immediately to a change in the preferences: no covariance matrix, no quadratic program, no numerical solver.

CES Utility: A Concentration Dial

For an elasticity of substitution $\eta > 0$, $\eta \neq 1$, the CES utility over the preferred set is $U_{\text{CES}}(\mathbf{n}) = \left(\sum_{i \in \mathcal{A}^+} \gamma_i \, n_i^{(\eta-1)/\eta}\right)^{\eta/(\eta-1)}$ (γ_i now a coefficient). Its maximizer is closed form, share counts proportional to the bang for the buck raised to η:

$$\boxed{n_i^\star = W_{\text{adj}} \, \frac{\left(\gamma_i / S_i(t)\right)^\eta}{\sum_{j \in \mathcal{A}^+} S_j(t) \left(\gamma_j / S_j(t)\right)^\eta} \quad (i \in \mathcal{A}^+)}$$

- $\eta \to 1$: the Cobb–Douglas allocation (its limit). $\eta \to \infty$: all of the preferred budget on the largest $\gamma_i / S_i(t)$ (when unique). $\eta \to 0^+$: equal share counts of every preferred asset.
- Engine's CES variant: $\eta(\xi_t) = \eta_{\min} + (\eta_{\max} - \eta_{\min})/(1 + |\xi_t|)$, $0 < \eta_{\min} \leq \eta_{\max}$ (0.5, 5 in the examples): concentrate when the market-state signal ξ_t (an EMA crossover, two frames on) is quiet, diversify when it is loud. Proofs, log-linear utility, and a share-denomination caveat: advanced notebook.

Preference Weights from the Single Index Model

From the SIM parameters (α_i, β_i) and two market inputs on day t, the recent market growth rate $\tilde{g}_{M,t}$ and a market-state signal ξ_t:

$$\boxed{\gamma_i(t) = \tanh\left(\frac{\alpha_i}{\beta_i^{\xi_t}} + \beta_i^{1-\xi_t}\,\tilde{g}_{M,t}\right)}$$

The $\tanh$ bounds the weight in $(-1, 1)$. Its argument is the SIM's expected growth of asset i in today's market, $\alpha_i + \beta_i \tilde{g}_{M,t}$ (inverse years), through a regime lens: it equals $(\alpha_i + \beta_i \tilde{g}_{M,t})/\beta_i^{\xi_t}$.

- Which assets belong in the basket: $\beta_i^{\xi_t} > 0$ and $\tanh$ keeps sign, so $\operatorname{sign}\gamma_i(t) = \operatorname{sign}(\alpha_i + \beta_i \tilde{g}_{M,t})$: preferred exactly when $\tilde{g}_{M,t} > -\alpha_i/\beta_i$. The signal never changes the classification; the recent market growth does.
- How much of each: the lens moves the magnitudes: bearish ($\xi_t > 0$) tilts toward low-beta names, bullish ($\xi_t < 0$) toward high-beta.

Preference Weights: The Basket, and Assumptions Stated

- As $\tilde{g}_{M,t}$ moves, assets cross between the sets: a strong market pulls in firms with a small negative α_i and a large β_i; a weak market pushes out everything but the strongest intercepts.
- If every $\gamma_i(t) \leq 0$, the model is saying this basket does not belong in this market: the allocator holds the floors plus cash, and an investor not content with cash should rethink which assets are in the basket (the ticker-picking methods of L12b and L13a; L7b's parameter updates move the thresholds too).
- Assumptions: (i) $\beta_i > 0$ (the course firms: 0.54 to 1.75, asserted; a non-positive beta needs a rule, e.g., assign to $\mathcal{A}^-$); (ii) the argument is an annualized growth rate read as a pure number at the one-year horizon; (iii) $\tilde{g}_{M,t}$ swings by about a unit per year, an order of magnitude more than the intercepts: that is what makes the basket respond, and why the schedule and rules decide how often it is acted on.

The Market Inputs

Both come from SPY with exponential moving averages, $\bar{S}_t = \omega S_t + (1-\omega)\bar{S}_{t-1}$, $\omega = 2/(L+1)$, window L days. With $L_{\text{short}} = 21$, $L_{\text{long}} = 63$, and a dimensionless gain $G > 0$, the signal is the scaled crossover:

$$\boxed{\xi_t = -G\left(\frac{\bar{S}_t^{\text{short}}}{\bar{S}_t^{\text{long}}} - 1\right)}$$

- Falling market (short EMA below long): $\xi_t > 0$, bearish; rising: $\xi_t < 0$, bullish; G sets $|\xi_t| \leq 1$ on 99% of training days ($G \approx 21$).
- $\tilde{g}_{M,t}$: the EMA (window $L_{\text{growth}} = 10$) of the daily market growth rates $g_{M,t} = \ln(S_t/S_{t-1})/\Delta t$. Both inputs on day t use prices through day $t-1$.

Example 1: CHEME-5660-L7a-Example-Utility-Allocator-Fall-2026.ipynb (first trading day of 2025: $\tilde{g}_M = -0.76$ per year, no firm preferred)

The Rebalancing Engine

The rebalancing engine: one pass per trading day

Signal from yesterday, trade at today's price, pay the cost, record, repeat

1. Observe
BAR t ARRIVES

- Execution prices $S_i(t)$, known at the close
- Signal from closes through $t-1$: EMAs $\bar{S}_{t-1}^{\text{short}}$, $\bar{S}_{t-1}^{\text{long}}$ give ξ_t
- SIM parameters α_i, β_i and g'_M, frozen today (L7b updates them)

2. Orient
PREFERENCES

Preference weight of firm i:

$$\gamma_i(t) = \tanh\left(\frac{\alpha_i}{\beta_i^{\xi_t}} + \beta_i^{1-\xi_t} g'_M\right)$$

- sign: $\gamma_i > 0$ is preferred ($\mathcal{A}^+$), otherwise non-preferred ($\mathcal{A}^-$)
- magnitude: share of the budget

3. Decide
TARGETS AND RULES

Cobb–Douglas targets on $\mathcal{A}^+$:

$$n_i^\star = \frac{\gamma_i}{\sum_{j \in \mathcal{A}^+} \gamma_j} \frac{W_{\text{adj}}}{S_i(t)}$$

- schedule: trade only if $t \in \mathcal{R}$
- turnover cap $\tau_{\max}$ scales the move
- drawdown limit $d_{\max}$: go to cash

4. Act
TRADE AND RECORD

- Trade Δn_i at $S_i(t)$
- Pay $C_t = c \sum_i |\Delta n_i| \, S_i(t)$ before the targets are sized
- $W_{\mathcal{P}}^{+}(t) = W_{\mathcal{P}}^{-}(t) - C_t$; update the cash
- Record positions, weights, cost

next bar $t+1$: cash accrues $e^{g_f \Delta t}$, mark to market at $S_i(t+1)$, check the drawdown breaker

A static allocation (L5b, L6b) is one pass on the decision date, then hold.

Every trading day is one pass: observe, orient, decide, act. A static allocation is one pass on the decision date, then nothing.

The Rebalancing Engine: The Account and the Timing

Let $n_{i,t}$ be the shares held after the close of day t; cash earns g_f. At the close of day t, before any trade, the book is worth:

$$W_{\mathcal{P}}^{-}(t) = \underbrace{\mathrm{cash}_{t-1}\, e^{g_f \Delta t}}_{\text{cash}} + \underbrace{\textstyle\sum_i n_{i,t-1}\, S_i(t)}_{\text{marked to market}}$$

Trading share deltas Δn_i at the close $S_i(t)$ moves gross notional $Q_t = \sum_i |\Delta n_i| S_i(t)$ and costs $C_t = c\, Q_t$ (c the cost rate). The cost comes out of the book and the positions are sized against what is left, the two solved together:

$$\boxed{W_{\mathcal{P}}^{+}(t) = W_{\mathcal{P}}^{-}(t) - C_t}$$

so $\mathrm{cash}_t + \sum_i n_{i,t} S_i(t) = W_{\mathcal{P}}^{+}(t)$ exactly and cash never goes negative. Next-bar execution: inputs and preferences for day t use prices through $t-1$; the trade executes at day t's close (market on close). A close-based signal traded at that close is look-ahead; Example 2 asserts the lag in code.

The Rebalancing Engine: Three Trigger Rules

Three rules bound the loop (else it trades daily, ignores a crash, or acts on every wiggle of the inputs):

- Reallocation schedule $\mathcal{R}$: trade only on days $t \in \mathcal{R}$ (daily, or every 21 days); hold otherwise.
- Turnover cap $\tau_{\max}$: with Δw_i the change of asset i's weight and $\Delta w_{\text{cash}} = -\sum_i \Delta w_i$, the one-way turnover $\tau_t = \frac{1}{2}(\sum_i |\Delta w_i| + |\Delta w_{\text{cash}}|)$ is the fraction of wealth that changes hands one way; a move with $\tau_t > \tau_{\max}$ is scaled down (the cost applies to the gross Q_t).
- Drawdown limit $d_{\max}$: a circuit breaker checked every day before the schedule; if $1 - W^{-}_{\mathcal{P}}(t) / \max_{s \le t} W^{-}_{\mathcal{P}}(s) > d_{\max}$, liquidate to cash (bypassing the cap), stay in cash n_{lock} more days, rebase the peak on the first scheduled trade after. It does not cap the recorded drawdown at $d_{\max}$.

They interact: all-cash to fully invested is a one-way turnover of one, so with $\tau_{\max} = 0.5$ the engine is only half invested until its next scheduled day.

Algorithm: Rebalancing Engine (Backtest), Setup

The engine is allocator-agnostic: it takes the prices and a target function, and our target functions wrap the allocators of this lecture around the lagged preference weights.

- **Given:** close prices $S_i(t)$, $t = 1, \dots, N$; the initial wealth $W_{\mathcal{P}}(0)$ in cash; the set $\mathcal{R}$; the rule parameters $(c, \tau_{\max}, d_{\max}, n_{\text{lock}})$.
- **Target function** $\mathbf{w}^{\star}(t, \text{state})$: returns risky weights ($w_i^{\star} \geq 0$, $\sum_i w_i^{\star} \leq 1$, remainder cash) from the day and the current state (pre-trade wealth, weights, shares, cash, and a copy of the day's execution prices).
- **Ours:** compute $\gamma_i(t)$ from the market inputs through $t-1$, allocate by Cobb–Douglas (or CES with $\eta(\xi_t)$) at the day's prices, return the allocation as weights; on an empty basket, floors plus cash.
- **Static baselines:** the same loop with a constant target and $\mathcal{R} = \{1\}$: same costs, same accounting.

Algorithm: Rebalancing Engine (Backtest), Daily Loop

For $t = 1, \dots, N$ (one bar at a time, under the timing rule):

1. Accrue cash by $e^{g_f \Delta t}$; mark to market at $S_i(t)$ to get $W^{-}_{\mathcal{P}}(t)$; update the running peak.
2. Breaker: if the drawdown exceeds $d_{\max}$ with risky positions held, liquidate at $S_i(t)$, pay c on the notional sold, lock n_{lock} more days, go to 5.
3. If $t \in \mathcal{R}$ and not locked: ask for $\mathbf{w}^{\star}$ (rebase the peak after a lock); form the proposed change of weights, scale to $\tau_{\max}$.
4. Solve cost and positions together on the post-cost budget; execute Δn_i at $S_i(t)$; pay $C_t = c\,Q_t$; update the cash.
5. Record wealth, cash, positions, weights, turnover, cost, flags: the histories every scorecard consumes.

Example 2: CHEME-5660-L7a-Example-Adaptive-Rebalancing-Scorecard-Fall-2026.ipynb

Scoring a Realized Path

Did the portfolio beat what the same dollars earn at the risk-free growth rate? For one outflow, one terminal value, constant g_f, horizon $T = N\Delta t$ years:

$$\boxed{\mathrm{NPV}(g_f, T) = \underbrace{-\,W_{\mathcal{P}}(0)}_{\text{investment}} + \underbrace{W_{\mathcal{P}}(T)\, e^{-g_f T}}_{\text{discounted terminal wealth}}}$$

positive means the run beat cash at g_f; the risk-free portfolio is the zero.

- Realized growth $\ln(W_{\mathcal{P}}(T)/W_{\mathcal{P}}(0))/T$; maximum drawdown $\max_t(1 - W_{\mathcal{P}}(t)/\max_{s\le t} W_{\mathcal{P}}(s))$.
- Realized Sharpe ratio: mean of the daily excess log growth over its standard deviation, times $\sqrt{1/\Delta t}$ (annualized, as in L6b).
- Turnover, cost, interventions: $\sum_t \tau_t$, $\sum_t C_t$, breaker firings.

One row per run, these columns. One path is one draw: what each policy did, not how likely it was; fail rates and tail quantiles need many paths (L7b).

Scoring a Realized Path: 2025 in One Table

Example 2: $\mathcal{R}$ every 21 days, $\tau_{\max} = 0.5$, $d_{\max} = 0.10$, $c = 5$ basis points, one engine for every run. The model emptied the basket on 57 of 250 days and preferred all thirteen firms on 152.

- Cobb–Douglas, monthly: 12 scheduled trades, floors plus cash on 50 days, no breaker firing; ended at 1.53 times initial wealth, drawdown 9.0% (rules off: 1.52).
- Cobb–Douglas, daily: 228 trades, one breaker firing, twenty times the cost; 1.40 (10.8%). CES with $\eta(\xi_t)$, monthly: 1.29 (9.2%).
- Buy and hold: tangent 1.28 (drawdown 31%), GMV 1.38 (13%), equal weights 1.59 (24%), SPY 1.17 (19%).

Read across, then against the baselines: who beat cash, what the basket selection did on the bad days, what a schedule cost. One year, one draw.

Example 2: CHEME-5660-L7a-Example-Adaptive-Rebalancing-Scorecard-Fall-2026.ipynb

Optional Advanced Material

Advanced: advanced/adaptive_utility/
CHEME-5660-L7a-Advanced-CES-Limits-Fall-2026.ipynb

Optional, not a prerequisite for L7b. Derives the CES closed form and proves its three limits (Cobb–Douglas at $\eta \to 1$, concentration at $\eta \to \infty$, equal share counts at $\eta \to 0^+$), reads the two limits of the elasticity rule $\eta(\xi_t)$, shows that log-linear utility has the Cobb–Douglas optimizer, and shows why the CES allocation over share counts depends on the price level of each share when $\eta \neq 1$.

Notebook: lectures/week-7/L7a/advanced, also linked from the lecture notebook.

Summary

Today: drift and fragile inputs, the Cobb–Douglas and CES allocators, preferences from the SIM and two market inputs, and an engine with an account, rules, and a scorecard.

Key takeaways

- Utility makes preferences explicit, Cobb–Douglas turns them into weights in closed form: sign puts an asset in or out of the basket, magnitude sets its share of the preferred budget; CES is a concentration dial.
- The SIM plus two market inputs supplies the preferences, the rules bound the response: recent market growth chooses the basket (empty means rethink it, or hold cash), the signal tilts the magnitudes; the rules decide when, how far, and whether to act.
- A realized-path scorecard is one draw: NPV against cash, drawdown, growth, Sharpe, turnover, cost say what happened; the distribution needs many paths.

Next, L7b: SIM parameters that update as data arrive, and an ensemble of futures.

Today: preferences computed every day, an allocation in closed form, and an engine that pays its own costs.